User Manual

Overview

This tool performs automated fairness testing on machine learning models. It evaluates whether predictions change when sensitive attributes (such as age, race, or gender) are modified.
The system compares two approaches:

- Random Search (baseline)
- Cluster-Guided Search (proposed method)

Step 1: Install Dependencies
Install required Python libraries:
pip install numpy pandas scikit-learn tensorflow matplotlib scipy joblib

Step 2: Prepare Files
Ensure the following files are available:
- dataset/processed_adult.csv
- dataset/processed_compas.csv
- DNN/model_processed_adult.h5
- DNN/scaler_adult.pkl
- DNN/model_processed_compas.h5
- DNN/scaler_compas.pkl

Step 3: Train Models (Optional)
If models are not available:
python train_model.py

This will train and save models and scalers.

Step 4: Run Experiments

Execute:
python experiments.py

What the Script Does

The script runs a full experimental suite:
1. Main Comparison
   Compares Random Search vs Cluster-Guided
2. K Variation
   Tests different cluster sizes (K = 5, 10, 15, 20)
3. Sensitive Feature Analysis
   Evaluates fairness across:
   - Age
   - Race
   - Gender
   - All combined
4. Budget Analysis
   Tests different query budgets (1000–5000)
5. COMPAS Evaluation
   Repeats experiments on a second dataset
6. Cross-Dataset Comparison
   Compares ADULT and COMPAS results

Output

All results are saved in the results/ folder:

CSV Files:

- exp1_adult_main.csv
- exp2_k_variation.csv
- exp3_sensitive_adult.csv
- exp4_budget_adult.csv
- exp5_compas_main.csv
- exp6_sensitive_compas.csv
- exp7_cross_dataset.csv

Figures:

- exp1_adult_main.png
- exp2_k_variation.png
- exp3_sensitive_adult.png
- exp4_budget_adult.png
- exp5_compas_main.png
- exp6_sensitive_compas.png
- exp7_cross_dataset.png

Key Parameters

- Discrimination threshold: 0.20
- Number of runs: 5
- Batch size: 256
- Default clusters: 20

Notes

- The model is treated as a black box
- Results may slightly vary due to randomness
- Execution time: ~10–15 minutes

This tool demonstrates an efficient and scalable fairness testing approach.